

History of Measurement

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Agenda

History of Measurement

The Earliest Linear Units

The Evolution of Linear Measurement

Measurement Systems

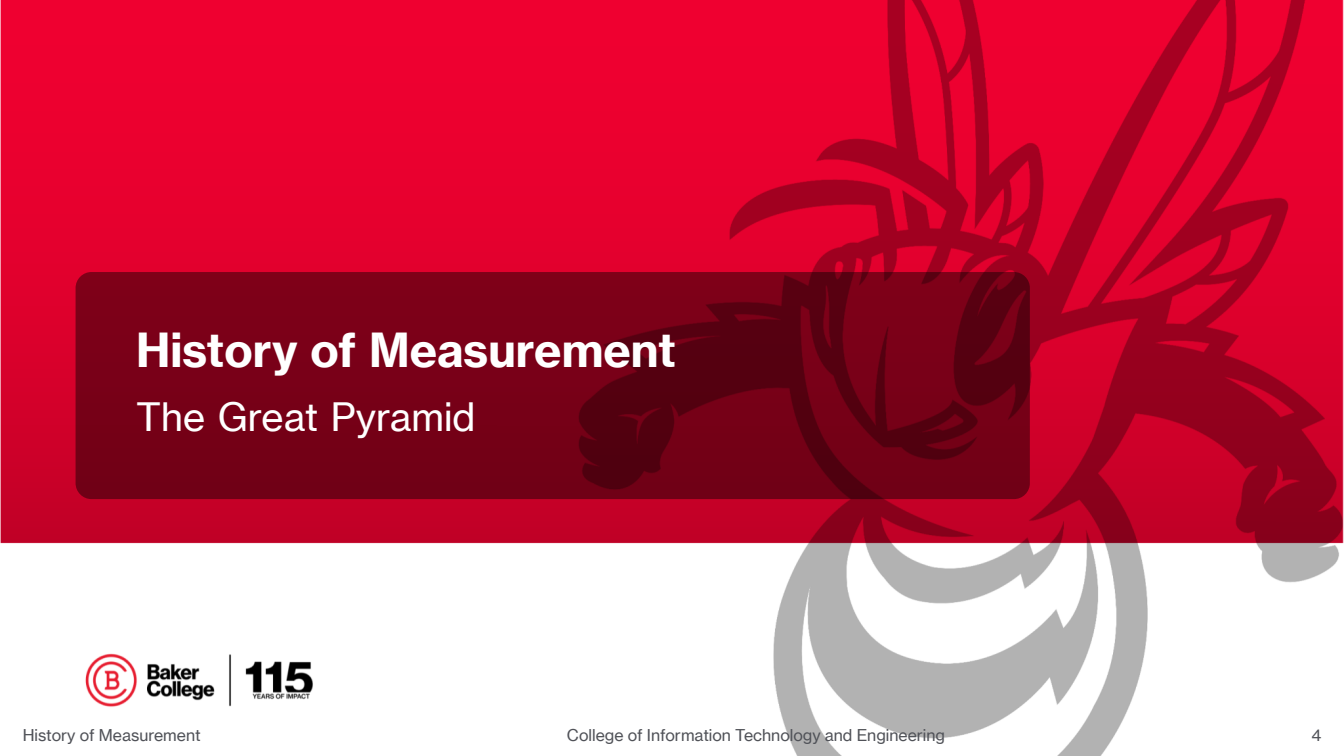
Measurement Terms

History of Measurement



History of Measurement

The Great Pyramid



Great Pyramid of Khufu (~2560 BC) (1 of 2)

- Built ~2560 B.C.
- Covers 13 acres
- ~755 foot long sides
- ~481 feet tall



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Great Pyramid of Khufu (~2560 BC) (2 of 2)

- All sides are equal ±0.6 in
- All angles are equal $\pm 0.03\overline{3}^{\circ}$



Editorial Note on the Pyramids

- We often hear “we don’t know how the pyramids were built”
- That doesn’t mean the builders had advanced tech, were time travelers, had alien intervention
- There are literally hundreds of possible, readily available methods–we just don’t know *which one* they used

History of Measurement

Eupalinian Aqueduct

Tunnel of Eupalinos (~500 BC) (1 of 2)

Herodotus, Greek historian, 484-425 BC:

> I have dwelt longer upon the history of the Samians than I should otherwise have done, because they are responsible for three of the greatest building and engineering feats in the Greek world: the first is a tunnel nearly a mile long, eight feet wide and eight feet high, driven clean through the base of a hill nine hundred feet in height. The whole length of it carries a second cutting thirty feet deep and three broad, along which water from an abundant source is led through pipes into the town.

Tunnel of Eupalinos (~500 BC) (2 of 2)

- 1,036 meters (3,399 ft) long
- Cut from opposite ends of the mountain
- Horizontal error: <1%
- Vertical error: <0.003% (around 1 inch)

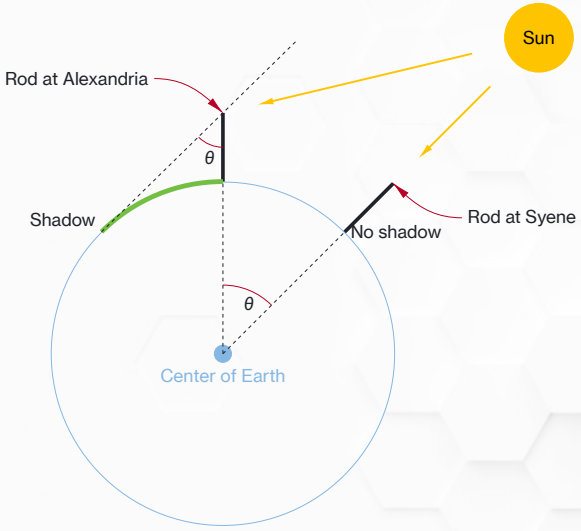


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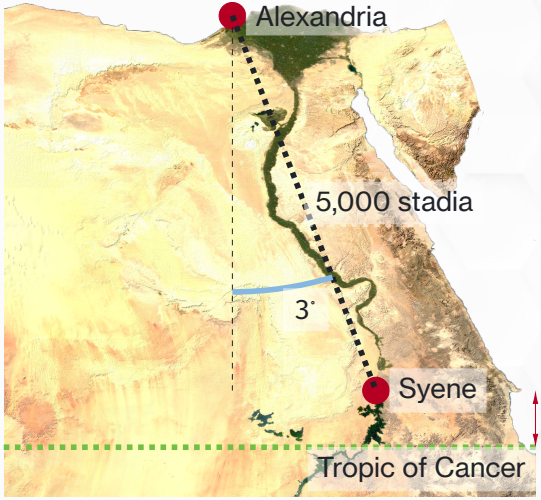
History of Measurement

Estimating Earth's Circumference

Eratosthenes' Earth Estimate (~240 BC) (1 of 2)



Eratosthenes' Earth Estimate (~240 BC)



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History of Measurement

Antikythera Mechanism

Antikythera Mechanism (~100 BC) (1 of 2)

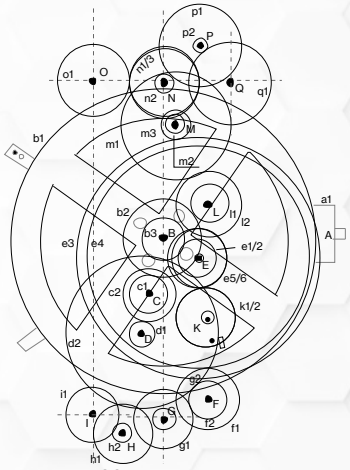
- Greek - discovered 1901 in a shipwreck
- Hand cranked computer
- Solar/lunar calendar, movements of stars/planets, schedule of Olympic games



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Antikythera Mechanism (~100 BC) (2 of 2)

- 30+ bronze gears
- Differential gear train modeling variable speed of the moon
- Pin and slot mechanisms
- History wouldn't see this kind of precision until 14th century clocks



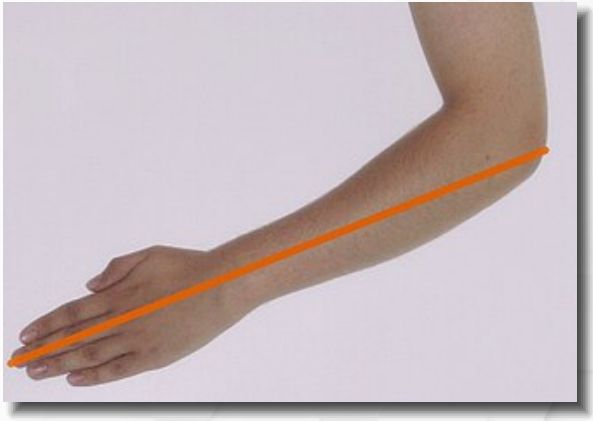
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The Earliest Linear Units



Cubit–The Earliest Unit

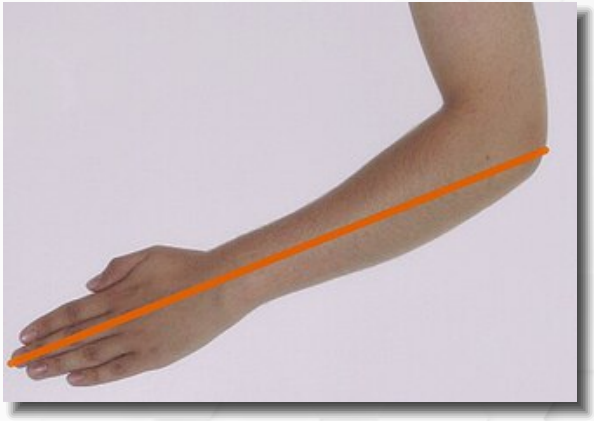
- The length from the elbow to the tip of the middle finger
- It may sound silly, but it's a measurement tool you always have with you



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Varying Measurements

- The big problem is that everyone's cubit is different
- There *were* standard cubits–one mainly used in Greece, the other in Egypt

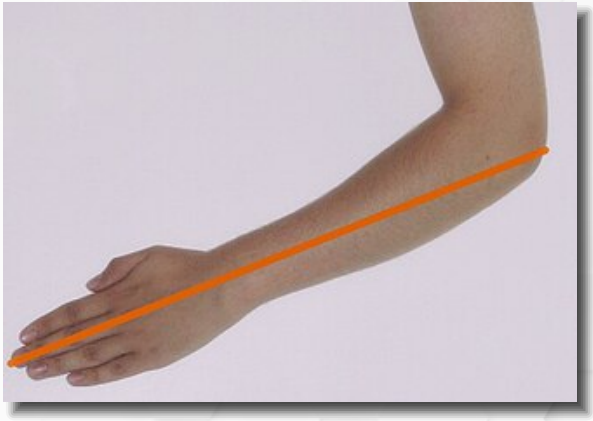


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A Tale of Two Cubits

Olympic Cubit ~18.24” (avg.)

Royal Egyptian Cubit ~20.62” (avg.)



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Measurement Standards

- They were smart enough to know that the measurement varied from person to person
- *Standards* are fixed size references to base other measurement tools on
- This is an Egyptian cubit standard, owned by an architect to the Pharoah



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Later Sub-units

Span 1/2 Olympic cubit (~9”)
Palm 1/6 Olympic cubit (~3”)
Digit 1/24 Olympic cubit (~3/4”)

The Evolution of Linear Measurement



The Foot and the Inch

- Over a bit of time the cubit start to be called a *foot*
- 1 foot was made up of 12 thumbnail widths (*unicae*)
- The unicae would eventually become the *inch*

The Timeline (1 of 2)

12th century 1 yard = Distance from the tip of King Henry I's nose to the end of his thumb

1324 1 inch = length of three barleycorns end to end

1588 Queen Elizabeth I commissioned a brass bar defined to be a standard yard

- Remained the standard until 1878

The Timeline (2 of 2)

16th century The foot ranged from 9-3/4” to 19”

1535 James Köbel, German mathematician and surveyor:

> Stand at the door of a church on a Sunday and bid 16 men to stop, tall ones and small ones, as they happen to pass out when the service is finished; then make them put their left feet one behind the other, and the length thus obtained shall be a right and lawful rood to measure and survey the land with, and the 16th part thereof shall be one right and lawful foot.

Weights and Standards Act of 1878

1 Yard Distance between two lines engraved in gold studs on a bronze bar as measured at 62˚F



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Why 62?

Why would 62°F matter?

Measurement Systems



The Metric System

- French scientists wanted permanent measurement system tied to something that didn't change

1793 Measured Earth's diameter, divided distance from equator to North Pole into 10 million parts = 1 metre

Current 1 meter = The distance travelled by light in a vacuum in 1/299,792,458 second

- The core tenet of the metric system is that units *must* be based on something that doesn't change

Metric Prefixes

Prefix	Fraction	Definition
milli	1/1,000	one-thousandth
centi	1/100	one-hundreth
deci	1/10	one-tenth
deka	10	ten
hekto	100	hundred
kilo	1,000	thousand

Metric Unit Equivalencies

Measurement	Equals
10mm	1cm
10cm	1dm
10dm	1m
10m	1dkm
10dkm	1hm
10hm	1km

The British Imperial System

Measurement	Equals
12 inches (in)	1 foot (ft)
3 feet (ft)	1 yard (yd)
5-1/2 yard (yd)	1 rod
40 rods	1 furlong
8 furlongs	1 mile (mi)

The American System (1 of 2)

1856 British government gives US two standard 1 yard bars, used as the US standard for years

1866 Congress legalized the metric system, 1 meter = 39.37 inches

- 1 yard = 3600/3937 meters
- 1 inch = 2.540005+ cm

The American System (2 of 2)

1889 US government given exact copy of 1 meter platinum bar to use as standard

1932 20˚C (68˚F) adopted as *standard measurement temperature*

1959 1 inch = 25.4mm exactly

Measurement Terms



Dimension

The measure of width, height, depth, or length in units of length of any object

Basic Dimension

The exact theoretical size from which all limiting variations are made

Tolerance

The permissible variation in the size of a part

Bilateral

1.000” ± 0.002”
1.00” + 0.001” / – 0.003”

Unilateral

1.000” + 0.000” / – 0.004”
1.000” + 0.004” / – 0.000”

All of these have a tolerance of 0.004”

Limits

The maximum and minimum dimensions obtained by applying the tolerances to the basic dimensions, the extreme dimensions beyond which the work cannot extend

- 2.251” ± 0.001”
- Lower limit = 2.251” - 0.001” = 2.250”
- Upper limit = 2.251” + 0.001” = 2.252”

Mean Dimension

Mean Dimension average of the high and low limits

- 0.250” +0.002”/-0.004”
- Lower limit = 0.250” - 0.004” = 0.246”
- Upper limit = 0.250” + 0.002” = 0.252”
- $\frac{0.252” + 0.246”}{2} = 0.249”$

Standard Size

A series of recognized or accepted sizes corresponding to various subdivisions of a recognized unit of length

- Bar stock – 1/64 inch, 1/32 inch, etc.
- Non-standard sizes are considered “odd sizes”
- Note that standard sizes may not match their dimensions

  Lumber 2x4 = 1.5x3.5
  Plumbing 3” pipe = 3” inner diameter
  Conduit 1” conduit = 1” outer diameter
- Standard sizes can be arbitrary letters or numbers

  Paper A4 = 210mm x 297mm

Questions?

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